High school · sequences

Telescoping sums

Progressive exercises

Exercise 1

Find $\sum[\frac{1}{k} - \frac{1}{k+1}]$ from $k = 3$ to 10.

Exercise 2

Express $\frac{1}{k(k+2)}$ as a difference of two fractions.

Exercise 3

Find $\sum(2k+1)$ from $k = 0$ to n using $(k+1)^2 - k^2$.

Exercise 4

If $S_n = \frac{n}{n+1}$, find the least integer $n \geq 1$ with $1 - S_n < 0.01$.